

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	30 % of CIA	Total Marks:	20 (Answer ANY ONE)
CO3 Statement:	Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)		
Student Name:	B.Nikitha	Reg. No.:	192373034
Section / Batch:	Slot A	Date:	7/08/2026

Code of Conduct

I _____B.Nikitha-192373034_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____B.Nikitha_____

Instructions

- Duration: 10 minutes

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

1. A TDMA system must support 480 users using 8 slots per carrier. Derive the number of carriers required, and then the total spectrum needed at 200 kHz per carrier. Present your derivation with intermediate steps clearly shown. Discuss how a network planner would use this exact calculation when sizing a new cell site for a projected subscriber base, and what safety margin might be added in practice.



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ITA0301 Mobile Computing Technical Presentation

Presented by
B.Nikitha(192373034)

Guided by
Dr Terrance Frederick Fernandez



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Given Data

- **Given**
- Total Users = **480**
- Slots per Carrier = **8**
- Bandwidth per Carrier = **200 kHz**
- **Formula**
- Number of Carriers = $\text{Total Users} / \text{Users Per Carriers}$



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- **Number of Carriers**
- **Step 1**
- Each carrier supports=8 Users
- Therefore,
- Number of Carriers= $480/8 = 60$
- **Step 2 – Calculate Total Spectrum**
- Bandwidth per Carrier = 200 kHz
- **Formula:** Total Spectrum = Number of Carriers $\times$ Bandwidth per Carrier
- $= 60 \times 200 \text{ kHz}$
- $= 12,000 \text{ kHz} = 12 \text{ MHz}$



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- **Practical Use in Network Planning**

- Estimate expected subscribers
- Calculate required carriers
- Determine required spectrum
- Check licensed spectrum availability
- Plan future network capacity

- **Benefits**

- Fewer call drops
- Less network congestion
- Better voice quality
- Efficient spectrum utilization



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Safety Margin

- **Why is a Safety Margin Added?**
- Users may increase in the future.
- More people use the network during peak hours.
- Festivals and special events increase network traffic.
- Extra capacity helps avoid call drops and congestion.
- **Example**
- Required carriers = **60**
- Add **20% safety margin**
- Extra carriers = **12**
- Total carriers = **72**



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Conclusion

To conclude, TDMA allows multiple users to share one carrier by dividing it into 8 time slots. For 480 users, we need 60 carriers and a total spectrum of 12 MHz. These calculations help network planners design new cell sites. In practice, a safety margin is added to handle future growth and peak traffic, ensuring reliable network performance.

Thank you



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Given Data

- Given
- Total Users = 480
- Slots per Carrier = 8
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- Formula
- Number of Carriers = $\frac{\text{Total Users}}{\text{Users Per Carriers}}$